

Experiment 3

Handling missing and inappropriate data in a dataset.

Code 1

```
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
print("Original Dataset:")
print(df)
print()

print("Duplicate rows check:")
print(df.duplicated())
print()

print("Dataset Info:")
df.info()
print()

df.drop_duplicates(inplace=True)
print("Dataset after removing duplicates:")
print(df)
print()
print("Number of rows after removing duplicates:", len(df))
print()

index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
print("Dataset after resetting index:")
print(df)
print()

df.drop(['Age_Group.1'], axis=1, inplace=True)
print("Dataset after dropping redundant column:")
print(df)
print()

df['CustomerID'] = df['CustomerID'].mask(df['CustomerID'] < 0)
df['Bill'] = df['Bill'].mask(df['Bill'] < 0)
df['EstimatedSalary'] = df['EstimatedSalary'].mask(df['EstimatedSalary'] < 0)
df['NoOfPax'] = df['NoOfPax'].mask((df['NoOfPax'] < 1) | (df['NoOfPax'] > 20))
print("Dataset after converting inappropriate values to NaN:")
print(df)
print()

print("Unique Age_Group values:", df.Age_Group.unique())
print("Unique Hotel values:", df.Hotel.unique())
print("Unique FoodPreference values:", df.FoodPreference.unique())
print()

df['Hotel'] = df.Hotel.replace(['Ibys'], 'Ibis')
df['FoodPreference'] = df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg')
df['FoodPreference'] = df.FoodPreference.replace(['non-Veg'], 'Non-Veg')
print("Unique Hotel values after cleaning:", df.Hotel.unique())
print("Unique FoodPreference values after cleaning:", df.FoodPreference.unique())
print()

df['EstimatedSalary'] = df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()))
df['NoOfPax'] = df.NoOfPax.fillna(round(df.NoOfPax.median()))
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()))
df['Bill'] = df.Bill.fillna(round(df.Bill.mean()))
print("Final Cleaned Dataset:")
print(df)
```

Output

```
Original Dataset:
   CustomerID  Age_Group  Rating(1-5)  Hotel FoodPreference  Bill \
0           1      20-25           4     Ibis             veg   1300
1           2      30-35           5  LemonTree          Non-Veg  2000
2           3      25-30           6    RedFox             Veg   1322
3           4      20-25          -1  LemonTree             Veg   1234
4           5       35+           3     Ibis      Vegetarian    989
```

5	6	35+	3	Ibys	Non-Veg	1909
6	7	35+	4	RedFox	Vegetarian	1000
7	8	20-25	7	LemonTree	Veg	2999
8	9	25-30	2	Ibis	Non-Veg	3456
9	9	25-30	2	Ibis	Non-Veg	3456
10	10	30-35	5	RedFox	non-Veg	-6755

	NoOfPax	EstimatedSalary	Age_Group.1
0	2	40000	20-25
1	3	59000	30-35
2	2	30000	25-30
3	2	120000	20-25
4	2	45000	35+
5	2	122220	35+
6	-1	21122	35+
7	-10	345673	20-25
8	3	-99999	25-30
9	3	-99999	25-30
10	4	87777	30-35

Duplicate rows check:

```

0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9     True
10   False
dtype: bool

```

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 11 entries, 0 to 10

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	11 non-null	int64
1	Age_Group	11 non-null	object
2	Rating(1-5)	11 non-null	int64
3	Hotel	11 non-null	object
4	FoodPreference	11 non-null	object
5	Bill	11 non-null	int64
6	NoOfPax	11 non-null	int64
7	EstimatedSalary	11 non-null	int64
8	Age_Group.1	11 non-null	object

dtypes: int64(5), object(4)

memory usage: 924.0+ bytes

Dataset after removing duplicates:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1	20-25	4	Ibis	veg	1300	
1	2	30-35	5	LemonTree	Non-Veg	2000	
2	3	25-30	6	RedFox	Veg	1322	
3	4	20-25	-1	LemonTree	Veg	1234	
4	5	35+	3	Ibis	Vegetarian	989	
5	6	35+	3	Ibys	Non-Veg	1909	
6	7	35+	4	RedFox	Vegetarian	1000	
7	8	20-25	7	LemonTree	Veg	2999	
8	9	25-30	2	Ibis	Non-Veg	3456	
10	10	30-35	5	RedFox	non-Veg	-6755	

	NoOfPax	EstimatedSalary	Age_Group.1
0	2	40000	20-25
1	3	59000	30-35
2	2	30000	25-30
3	2	120000	20-25
4	2	45000	35+
5	2	122220	35+
6	-1	21122	35+
7	-10	345673	20-25
8	3	-99999	25-30
10	4	87777	30-35

Number of rows after removing duplicates: 10

Dataset after resetting index:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	\
0	1	20-25	4	Ibis	veg	1300	2	
1	2	30-35	5	LemonTree	Non-Veg	2000	3	
2	3	25-30	6	RedFox	Veg	1322	2	

3	4	20-25	-1	LemonTree	Veg	1234	2
4	5	35+	3	Ibis	Vegetarian	989	2
5	6	35+	3	Ibys	Non-Veg	1909	2
6	7	35+	4	RedFox	Vegetarian	1000	-1
7	8	20-25	7	LemonTree	Veg	2999	-10
8	9	25-30	2	Ibis	Non-Veg	3456	3
9	10	30-35	5	RedFox	non-Veg	-6755	4

EstimatedSalary Age_Group.1		
0	40000	20-25
1	59000	30-35
2	30000	25-30
3	120000	20-25
4	45000	35+
5	122220	35+
6	21122	35+
7	345673	20-25
8	-99999	25-30
9	87777	30-35

Dataset after dropping redundant column:

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	\
0	1	20-25	4	Ibis	veg	1300	2
1	2	30-35	5	LemonTree	Non-Veg	2000	3
2	3	25-30	6	RedFox	Veg	1322	2
3	4	20-25	-1	LemonTree	Veg	1234	2
4	5	35+	3	Ibis	Vegetarian	989	2
5	6	35+	3	Ibys	Non-Veg	1909	2
6	7	35+	4	RedFox	Vegetarian	1000	-1
7	8	20-25	7	LemonTree	Veg	2999	-10
8	9	25-30	2	Ibis	Non-Veg	3456	3
9	10	30-35	5	RedFox	non-Veg	-6755	4

EstimatedSalary	
0	40000
1	59000
2	30000
3	120000
4	45000
5	122220
6	21122
7	345673
8	-99999
9	87777

Dataset after converting inappropriate values to NaN:

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1	20-25	4	Ibis	veg	1300.0
1	2	30-35	5	LemonTree	Non-Veg	2000.0
2	3	25-30	6	RedFox	Veg	1322.0
3	4	20-25	-1	LemonTree	Veg	1234.0
4	5	35+	3	Ibis	Vegetarian	989.0
5	6	35+	3	Ibys	Non-Veg	1909.0
6	7	35+	4	RedFox	Vegetarian	1000.0
7	8	20-25	7	LemonTree	Veg	2999.0
8	9	25-30	2	Ibis	Non-Veg	3456.0
9	10	30-35	5	RedFox	non-Veg	NaN

NoOfPax	EstimatedSalary
0	2.0
1	3.0
2	2.0
3	2.0
4	2.0
5	2.0
6	NaN
7	NaN
8	3.0
9	4.0

Unique Age_Group values: ['20-25' '30-35' '25-30' '35+']
Unique Hotel values: ['Ibis' 'LemonTree' 'RedFox' 'Ibys']
Unique FoodPreference values: ['veg' 'Non-Veg' 'Veg' 'Vegetarian' 'non-Veg']

Unique Hotel values after cleaning: ['Ibis' 'LemonTree' 'RedFox']
Unique FoodPreference values after cleaning: ['Veg' 'Non-Veg']

Final Cleaned Dataset:

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	\
0	1	20-25	4	Ibis	Veg	1300.0
1	2	30-35	5	LemonTree	Non-Veg	2000.0
2	3	25-30	6	RedFox	Veg	1322.0
3	4	20-25	-1	LemonTree	Veg	1234.0
4	5	35+	3	Ibis	Veg	989.0

5	6	35+	3	Ibis	Non-Veg	1909.0
6	7	35+	4	RedFox	Veg	1000.0
7	8	20-25	7	LemonTree	Veg	2999.0
8	9	25-30	2	Ibis	Non-Veg	3456.0
9	10	30-35	5	RedFox	Non-Veg	1801.0

	NoOfPax	EstimatedSalary
0	2.0	40000.0
1	3.0	59000.0
2	2.0	30000.0
3	2.0	120000.0
4	2.0	45000.0
5	2.0	122220.0
6	2.0	21122.0
7	2.0	345673.0
8	3.0	96755.0
9	4.0	87777.0